AICTE – BharatCares – IBM SkillsBuild Internship 2026

PHISHGUARD//NEXUS: FINAL CAPSTONE DOSSIER

Applied AI & Machine Learning Track | Complete Submission Package

Student Name:	Sahil Rai
Internship ID:	IBMUEML002432
Institution:	Lovely Professional University (LPU), Punjab
Live Cloud URL:	https://phishguard-nexus.vercel.app/ (tinyurl.com/pg-nexus)
GitHub Repo:	https://github.com/rootbysahil/phishguard-nexus

1. Executive Summary & UN SDG Alignment

PHISHGUARD//NEXUS is an Explainable AI cybersecurity platform engineered to detect and dismantle zero-day phishing across URLs, emails, and SMS without external network lookups. Operating on 24 pure-Python lexical signals and a calibrated Logistic Regression classifier (ROC-AUC: 1.000, F2: 0.9989), it integrates directional SHAP feature attributions, IBM watsonx / Granite LLM multilingual advisories, and direct integration with India's National Cybercrime infrastructure (Helpline 1930 / cybercrime.gov.in). Aligned with UN SDGs 9 (Target 9.c) and 16 (Target 16.4).

2. The Lean Canvas (9-Box Matrix)

<table>
<tr>
<td rowspan="2">1. PROBLEM
• Zero-day phishing expires <4h.
• Black-box AI causes mistrust.
• Scanners leak victim IP.
Alternatives: VirusTotal, Google Safe Browsing.</td>
<td>4. SOLUTION
• 24 lexical features (0 network calls).
• Calibrated ML (ROC-AUC 1.000, F2 0.9989).
• Directional SHAP XAI.
• Multilingual watsonx advisories.
• Automated forensic PDF reports.</td>
<td rowspan="2">3. UNIQUE VALUE PROP
"Zero-latency forensic defense that explains WHY a link is dangerous and WHAT TO DO in regional languages."
High-Level Concept:
VirusTotal + IBM XAI + 1930 Helpline.</td>
<td>9. UNFAIR ADVANTAGE
• Zero-leakage domain-grouped split.
• Privacy-first zero-call architecture.
• Native Indian Cybercrime 1930 escalation.</td>
<td rowspan="2">2. CUSTOMER SEGMENTS
• Indian UPI banking users.
• Elderly citizens vulnerable to KYC scams.
• MSMEs without dedicated SOCs.
Early Adopters: Students & fintech users.</td>
</tr>
<tr>
<td>8. KEY METRICS
• ROC-AUC: 1.0000 | F2: 0.9989
• <15ms latency | Total scans logged</td>
<td>5. CHANNELS
• Vercel Web App (tinyurl.com/pg-nexus)
• Mobile QR PWA | IBM SkillsBuild Showcase</td>
</tr>
<tr>
<td colspan="2">7. COST STRUCTURE
• Vercel Serverless ($0 Hobby)
• Edge SQLite | IBM watsonx token inference</td>
<td colspan="3">6. REVENUE STREAMS
• Free Community Tier | Enterprise B2B API
• Institutional Forensic Audits | Govt Grants</td>
</tr>
</table>

3. Technical Architecture & Verified Performance

• **24 Lexical Signals:** Evaluated in <15ms with 0 external network requests to prevent victim IP leakage.
• **Champion Classifier:** Calibrated Logistic Regression (Platt scaling) trained on domain-isolated splits (GroupShuffleSplit).
• **ROC-AUC:** 1.0000 | **F2 Score:** 0.9989 (95% CI: [0.997, 1.000]) | **Recall:** 1.0000 | **Precision:** 0.9989.
• **Directional SHAP (XAI):** LinearExplainer assigns feature-level weights informing users of exact risk triggers.
• **IBM watsonx.ai Integration:** Synthesizes forensic findings into plain language and translates to English, Hindi, and Punjabi.
• **Cryptographic PDF Audits:** Generates downloadable forensic sheets stamped with SHA-256 evidence hashes.

4. SkillsBuild 6-Week Masterclass Milestones

Week	SkillsBuild Masterclass	Project Accomplishment Executed
Week 1	Kickstart ML & Tools (VS Code, BOB)	Environment setup, dataset collation & API contracts.
Week 2	Supervised Learning & IBM watsonx	24-feature extractor, baseline heuristics & classifier benchmarking.
Week 3	Unsupervised Patterns & Deep Learning	Shannon entropy, homoglyph clustering & GroupShuffle validation.
Week 4	Intelligent Agents & Lifelong Skills	SHAP XAI engine, watsonx advisory prompt orchestration.
Week 5	Project Completion & Q&A;	FastAPI backend, Cyberpunk UI & ReportLab PDF engine.
Week 6	Project Submission (PPT & Code)	Live Vercel cloud deployment, automated test verification (8/8).

5. Student Undertaking & Submission Declaration

I hereby declare that this project, **PHISHGUARD//NEXUS**, has been developed as my final capstone deliverable for the 6-week virtual IBM SkillsBuild Machine Learning and Applied AI Internship (2026) in collaboration with AICTE and BharatCares. The platform is verified, passing all unit and integration tests (8/8), and live at **https://phishguard-nexus.vercel.app/** (shortlink: **tinyurl.com/pg-nexus**).

Student Signature: Sahil Rai **Date of Submission:** 13/10/2026